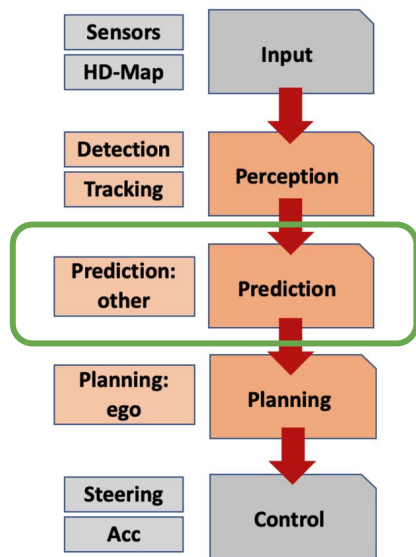




# LAformer Explainability

Sri Nitchith Akula

# TLDR;



Self-Driving Stack

**Motivation:** Trajectory prediction models for autonomous driving are black boxes and understanding why a model predicts a certain trajectory is critical for safety and trust.

**Approach:** Apply perturbation-based explainability (agent ablation, lane ablation, temporal occlusion) to LAformer, a lane-aware transformer.

## Key Results:

- Lanes are the dominant feature (removing lanes shifts FDE more than compared to removing agents)
- Nearby vehicles influence prediction far more than distant pedestrians
- Built an interactive Gradio demo for real-time "what-if" exploration

Code: <https://github.com/nitchith/LAformer-Explainability>



# Problem Statement

## Why explainability for trajectory prediction?

- Autonomous vehicles rely on trajectory prediction to anticipate other road users
- Modern models (GNNs, Transformers) achieve strong accuracy but are opaque
- For safety-critical deployment, we need to understand:

| Question                                    | Why it matters                        |
|---------------------------------------------|---------------------------------------|
| Which agents influence the prediction?      | Identify safety-critical interactions |
| How important are lanes vs. social context? | Validate architectural design choices |
| Which moments in history matter most?       | Understand temporal dependencies      |



# Literature Review: Trajectory Prediction

| Model           | Year        | Venue       | Key Idea                                              |
|-----------------|-------------|-------------|-------------------------------------------------------|
| Social LSTM     | 2016        | CVPR        | Learned social pooling over neighbors                 |
| Social GAN      | 2018        | CVPR        | Adversarial multi-modal prediction                    |
| VectorNet       | 2020        | CVPR        | Polyline-based vectorized scene encoding              |
| Trajectron++    | 2020        | ECCV        | Conditional VAE + interaction graph                   |
| QCNet           | 2023        | CVPR        | Query-centric encoding + two-stage decoder            |
| Wayformer       | 2023        | ICRA        | Factored attention over agents and road               |
| MTR             | 2023        | NeurIPS     | Motion Transformer with intention points              |
| <b>LAformer</b> | <b>2024</b> | <b>CVPR</b> | <b>Lane-aware Transformer with lane scoring</b>       |
| HPNet           | 2024        | CVPR        | Historical prediction attention for stability         |
| GoalFlow        | 2025        | CVPR        | Goal-driven flow matching for multimodal trajectories |

**Trend:** Increasing use of map/lane features alongside social context, with Transformer architectures dominating recent SOTA.



# Literature Review: Explainability in ML

Common post-hoc methods:

- LIME (Ribeiro et al., 2016) => local surrogate models
- SHAP (Lundberg & Lee, 2017) => Shapley value attribution
- Gradient-based => Saliency maps, GradCAM

Why perturbation-based ablation for trajectory prediction?

- Input is a structured graph (polylines), not images => Might be hard to use gradient methods
- Helps with answering: What happens if this agent/lane wasn't there?
- Model-agnostic i.e works on any prediction model without architecture changes
- Computationally feasible: LAformer inference is ~0.2s per sample on CPU

# LAformer Architecture

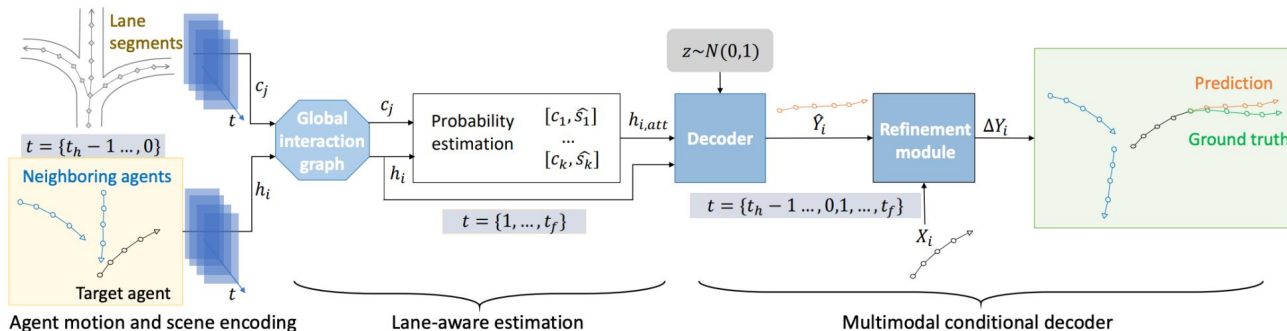


Figure 2. The LAformer framework takes vectorized trajectories and HD map lane segments as input. The agent motion and scene encodings are represented as  $h_i$  and  $c_j$ , respectively, and are later fused by the attention-based global interaction graph. The decoder then takes the target agent's past trajectory  $h_i$ , the updated motion information aligned with the candidate lane information  $h_{i,att}$  from the lane-aware estimation module, and a random latent variable  $z$  as inputs, and predicts the future trajectory  $\hat{Y}_i$ . The refinement module further reduces the predicted offset  $\Delta Y$  to improve prediction accuracy.



# Approach

## 1. Neighbor-Agent Ablation (Leave-One-Out)

Remove each agent's polyline individually, measure ADE/FDE shift from baseline.

## 2. Group Ablation

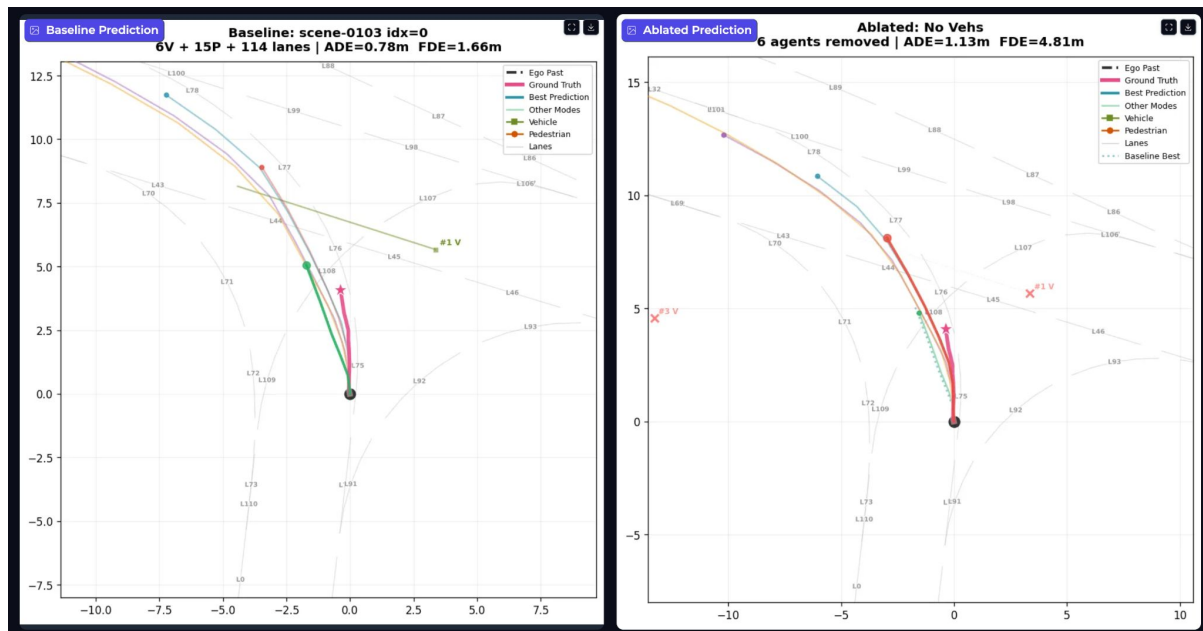
Remove entire categories and compare shifts:

- No/All Lanes
- No/All Vehicles
- No/All Pedestrians
- Only Ego vehicle

## 3. Temporal Occlusion

Mask each ego history timestep individually to find which moments matter most.

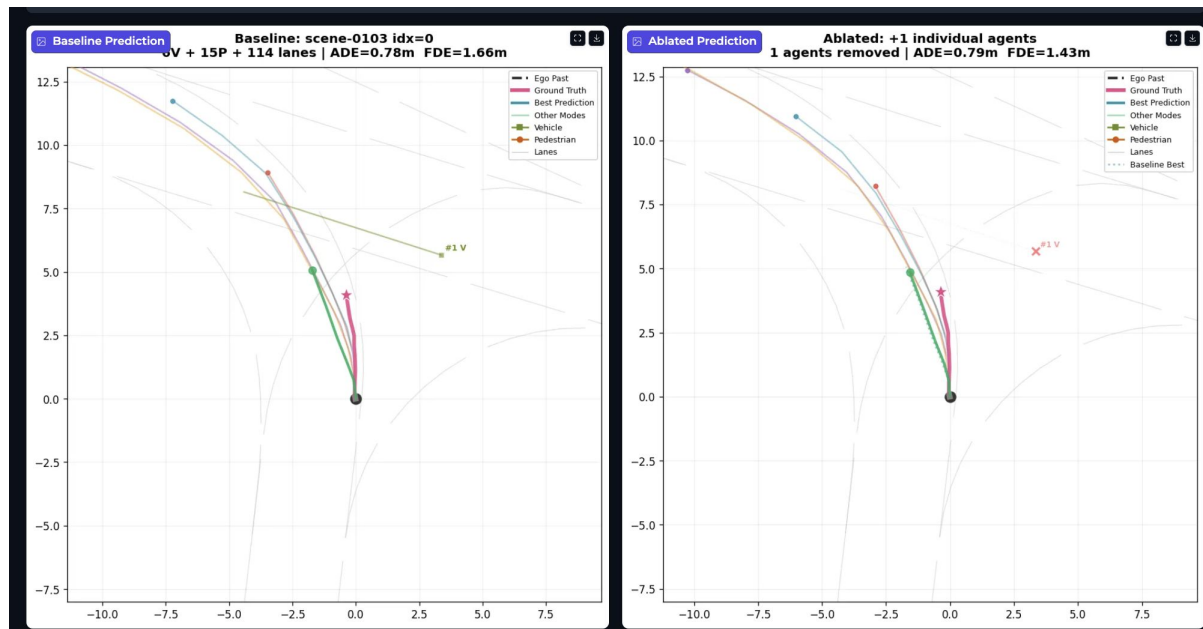
# Results (All Vehicle Masking)



By removing all the other vehicles in the scene, we can see that the most-likely prediction is moving further.

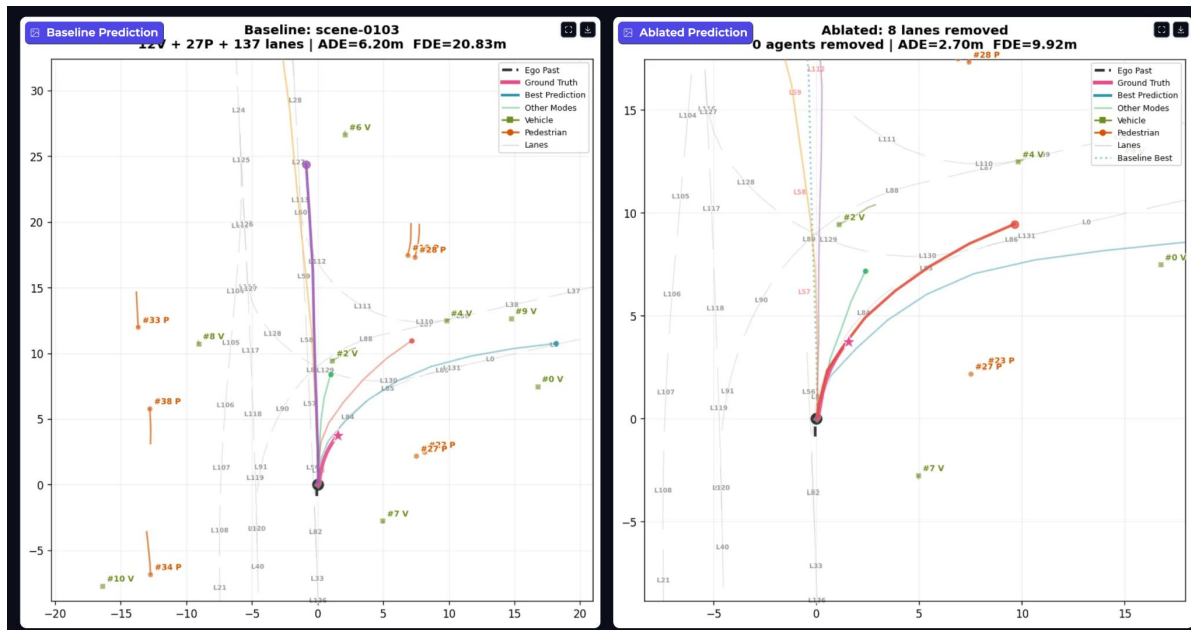
This can indicate that the model might be yielding for the vehicle #1 – but that's not the case as seen in the next slide

# Results (Masking Veh #1)



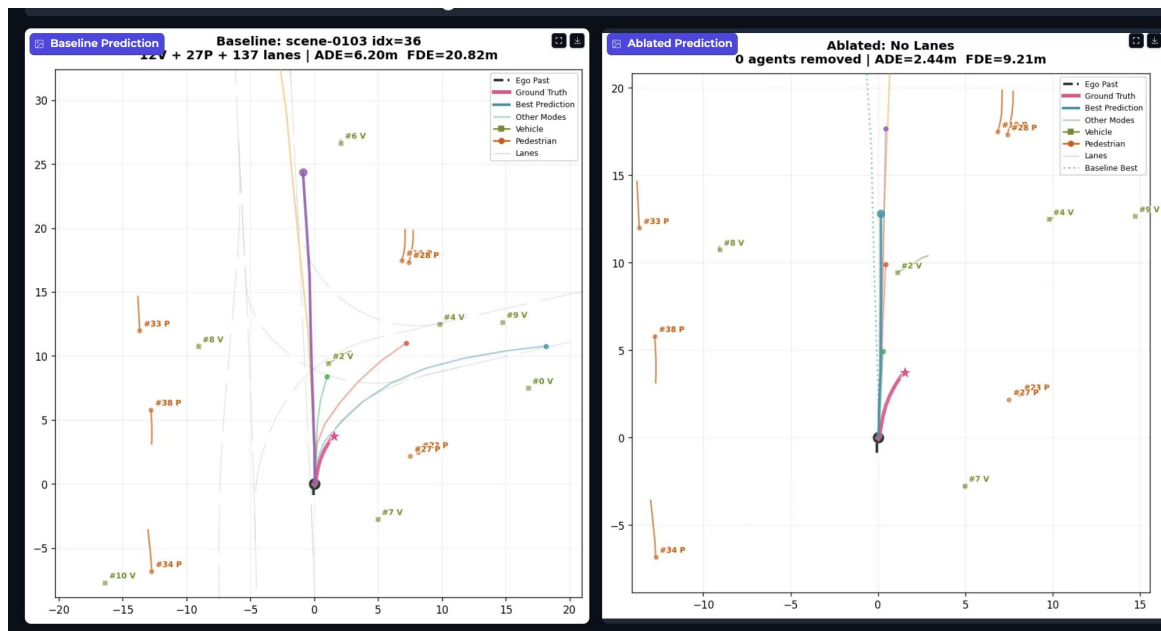
Just making for agent #1, we don't see the best prediction to be proceeding

# Results (Partial Lane Masking)



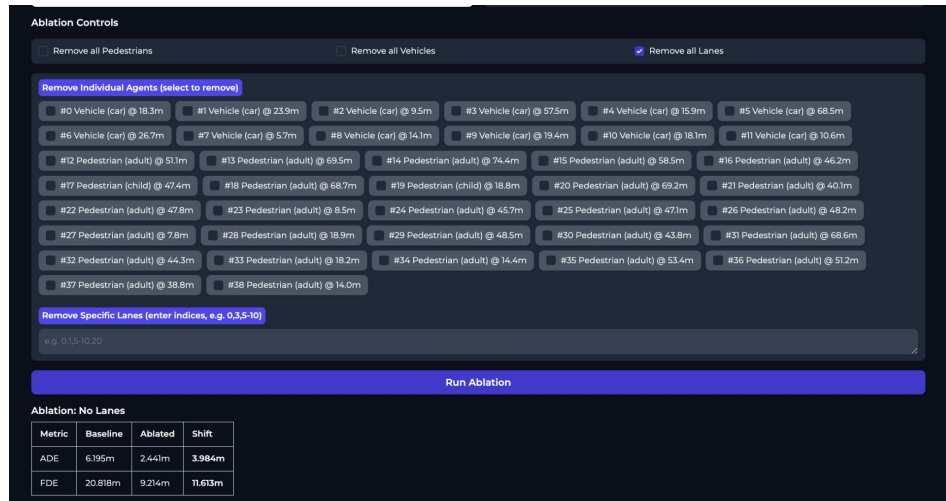
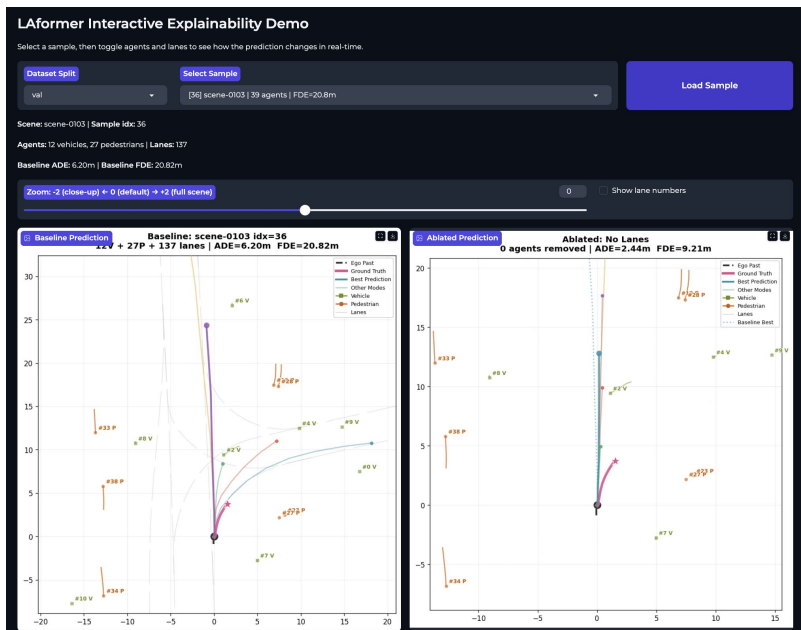
By removing the straight going lanes, the most-likely predictions is now turing

# Results (Full Lane Masking)



By removing all the lanes, the model seems to rely more on its own history to go straight

# Demo UI Screenshots





# Demo

Youtube link: [https://youtu.be/mDdcummOW\\_s](https://youtu.be/mDdcummOW_s)

Code: <https://github.com/nitchith/LAformer-Explainability>



## Conclusion

- Case study examples have shown interesting prediction changes due to the input changes
- There are many examples where the predictions are bad without any agent interaction.
- Explainability analysis is only as meaningful as the model's baseline accuracy i.e, if predictions aren't lane-consistent, ablation results become harder to interpret.
- Apply this framework to models with stronger agent interaction and lane conditioning (e.g., MTR, QCNet) for more interpretable ablation results



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